

Likelihood II

Today's agenda:

- Check-in
- Confidence Intervals

Check-in

How are things going?

Anything that could be better?

Additional resources that could be added?

More assignments or examples?

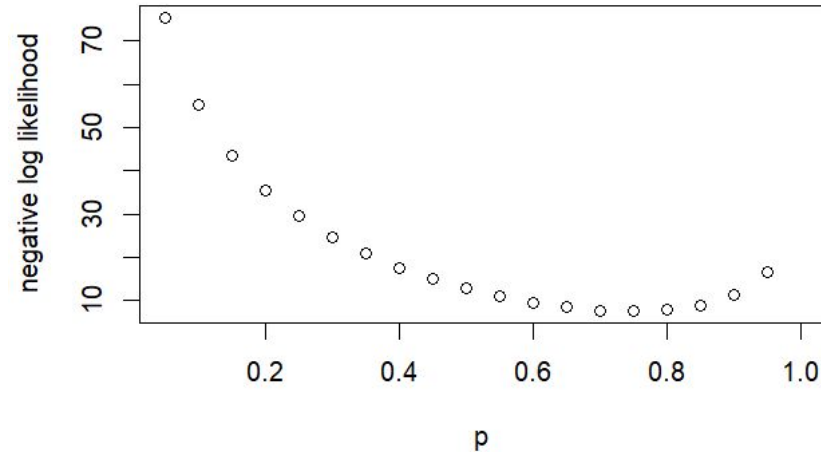
Last time

- Fitting functions with multiple parameters:

```
binomNLL2 <- function(params, N, k){  
  a = params[1]  
  h = params[2]  
  pred_prob = a / (1+a*h*N)  
  -sum(dbinom(x = k, size = N, prob = pred_prob, log = TRUE))  
}
```

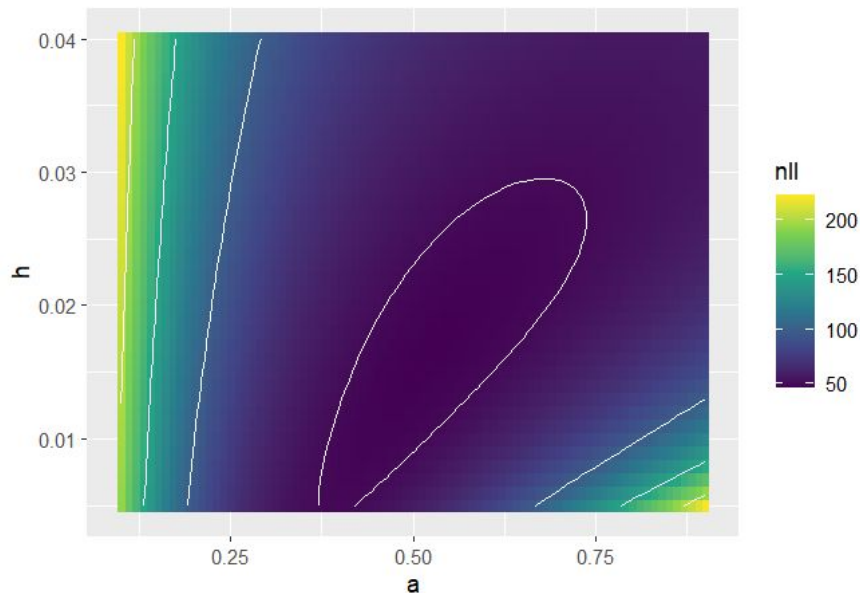
The time before last

- Likelihood curves:
 - Likelihood depends on a parameter value



How do we handle likelihood plots in multiple dimensions?

- Likelihood surfaces (instead of curves)
- Essentially, you evaluate the likelihood at every combination of parameters



Other options

What if we have more than 2 parameters?

What if we want to visualize one parameter at a time? (surfaces hard to visualize)

Options:

- Try plotting in more than 2 dimensions (not advised)
- Take a slice through the data
- Calculate likelihood profiles

Slices

- Pick a focal parameter and a range of values to evaluate it over
- Set other parameters to some value (e.g., their max likelihood estimates)
- Calculate likelihood
- May be useful, but are misleading
 - Parameters trade-off, so the parameter combinations you try aren't always reasonable

Slices: Worked example pt 1

```
data("ReedfrogFuncresp")

binomNLL2 <- function(params, N, k){
  a = params[1]
  h = params[2]
  pred_prob = a / (1+a*h*N)

  -sum(dbinom(x = k,size = N, prob = pred_prob,log = TRUE))
}

parnames(binomNLL2) <- c("a","h") # quirk of mle2: need to set the parm names

m2 <- mle2(minuslogl = binomNLL2,
           start = c(a=.5,h=2),
           data = list(k = ReedfrogFuncresp$Killed,
                       N = ReedfrogFuncresp$Initial))
```


Slices: Worked example pt 2

```
opt_a <- opt2$par["a"] # get the optimal parameter value for a then do a slice for h

h_vec <- seq(from = 0.01,to=0.04,by=0.005) # pick a sequence to optimize over

# we're going to use mle2 because it makes it easy to fix certain parameters

h_nll <- numeric(length(h_vec)) # this will hold our output

for(i in 1:length(h_vec)){

  m2 <- mle2(minuslogl = binomNLL2,
             start = c(a=.5,h=2),
             data = list(k = ReedfrogFuncresp$Killed,
                         N = ReedfrogFuncresp$Initial),
             fixed = list(a = as.numeric(opt_a),
                         h = h_vec[i]))

  h_nll[i] <- m2@min

}
```

Slices: Worked example pt 3

Plot your slice ($x = h_vec$, $y = h_nll$)

How does the minimal value on your slice compare with the the optimization?

Slices: Your turn

- You've done a slice for h (keeping a fixed at the optimized value)
- Now, modify the code to do a slice for a (keeping h fixed)

Profiles

- Pick a focal parameter and a range of values to evaluate it over
 - Allow other parameters to be optimized (*different from slices)
 - Calculate likelihood
-
- Still not a complete picture, but represents actual parts of the surface

Profiles

How might we modify the slice code to calculate a profile?

Profiles

How might we modify the slice code to calculate a profile?

- Since we used `mle2`, we can just remove “a” from the fixed arguments

Profiles

- Modify your previous code to:
 - Leave out “a” from the fixed terms
 - Use a different object to store the new NLL values (e.g., `h_nll2`)
 - Store the “a” values that are being chosen
- Plot 1:
 - Your profile for h (as before, the x-axis is the value of h, and y is the NLL)
 - Add points for the slice you took over the same values in another color
- Plot 2:
 - For your profile, plot a vs h

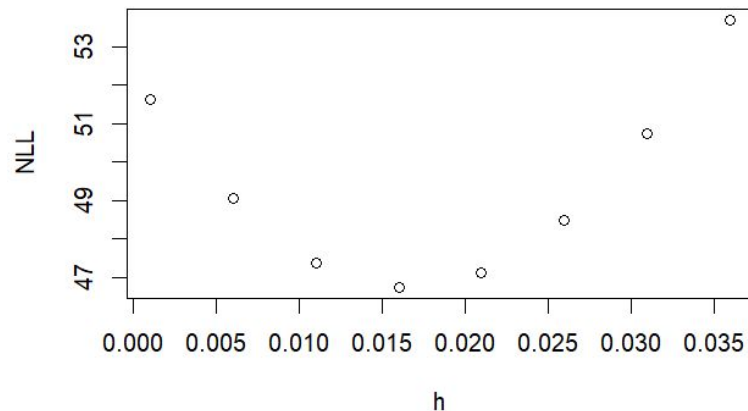
Profiles

From your plots:

- Which has the lower value for NLL?
- What does the plot of a vs h tell us about why this is?

Parameter confidence limits

- Parameters close to the best might be feasible
- Those farther away are probably not
- Commonly use likelihood-ratio tests



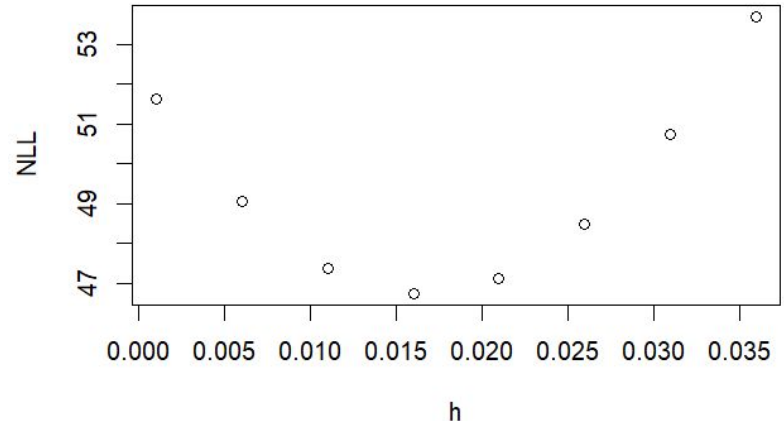
Likelihood Ratio test: General Idea

You can compare the likelihood of two combinations of parameters

$$\text{Deviance} = -2\log(\text{best model likelihood} / \text{restricted model likelihood})$$

Deviance follows a χ^2 distribution with r degrees of freedom

(r = number of fixed factors in the restricted model)



Likelihood Ratio Test in practice

- You can do an LRT manually (see pg 193-194 in your books)
- but R makes it easier:

```
m5 <- mle2(minuslogl = binomNLL2,  
           start = c(a=.5,h=2),  
           data = list(k = ReedfrogFuncresp$Killed,  
                       N = ReedfrogFuncresp$Initial))
```

```
profile(m5)
```

Likelihood Ratio Test in practice

Likelihood profile:

 Difference in NLL from best

\$a

	z	par.vals.a	par.vals.h
1	-4.2049805	0.294888378	-0.002924909
2	-3.1540453	0.341234208	0.002591865
3	-2.2328584	0.387580038	0.007020138
4	-1.4114294	0.433925867	0.010708266
5	-0.6687860	0.480271697	0.013875313
6	0.0000000	0.526617527	0.016648015
7	0.6370924	0.572963357	0.019132317

 Best model

Likelihood Ratio Test in practice

Getting the confidence interval:

```
confint(profile(m5), level = 0.95)
```

Or equivalently

```
confint(m5, level = 0.95)
```

Try these and see what you get

Likelihood Ratio Test in practice

Plotting the profile

```
plot(profile(m5))
```

Likelihood Ratio Test in practice

Adding the CI to our profile plot:

```
ci_m5 <- confint(m5, level = 0.95)
plot(x = h_vec, y = h_nll2, ylab = "NLL", xlab = "h")
abline(v = ci_m5["h",]["2.5 %"])
abline(v = ci_m5["h",]["97.5 %"])
```

Before next class:

- 6.6 (6.6.3 optional)